# Why the $N{=}7$ Chebyshev CRRA fixed point floors at $\|F\|_\infty \sim 3\times10^{-3}$, and what to do about it

Fixed-Point Factory — diagnostic note

June 2026

**Abstract**

We asked all five available solvers (two Picard variants, two Anderson-accelerated variants, and a Levenberg–Marquardt minimiser) to drive the K=3 CRRA Hellwig operator to machine precision in the $S_3$-symmetric subspace at $N{=}7$, $\tau{=}\gamma{=}1$. *All five floor at* $\|F\|_\infty \in [1.2, 5.3]\times 10^{-3}$. A direct linearisation of the operator at the best LM iterate gives spectral radius $\rho(J) = 4.58$ with four eigenvalues outside the unit circle: the operator *repels* in the symmetric subspace, so no Picard / Anderson / Newton scheme can converge there. Levenberg–Marquardt, which minimises $\|F\|_2^2$ rather than solving $F = 0$, settles at $\|F\|_\infty = 2.70\times10^{-3}$ — and that is the genuine minimum of the residual, not a tolerance artefact. We trace the non-vanishing residual to a known structural feature of the contour-extraction operator: the integrand jumps as the conjectured price polynomial gains or loses roots inside the integration box. The closed-form rank-1 and 1D-in-$T$ operators, which never extract roots, hit machine $\varepsilon$ and $4{\times}10^{-6}$ respectively, confirming the issue is the root-finding step itself. We then describe an "Option 3" regularisation — explicit boundary corrections with smooth cutoff weights — that is the next thing to try.

## Contents

## 1 Problem statement

The K=3 CRRA REE operator $\Phi$ in $S_3$-symmetric Lobatto–Chebyshev coordinates at $N$=7 has 40 free degrees of freedom in the symmetric subspace (down from $7^3 = 343$ in the full cube). Given a conjecture $P(u_1, u_2, u_3)$, one step of $\Phi$ does:

1. For each Lobatto signal $u_k$, extract the level set $\{(u_a, u_b) : P(u_a, u_b, u_k) = p\}$ by Chebyshev root-finding;

2. integrate $f_v\, f_v/\|\nabla P\|$ along that curve to get co-area weights $A_v(p, u_k)$;

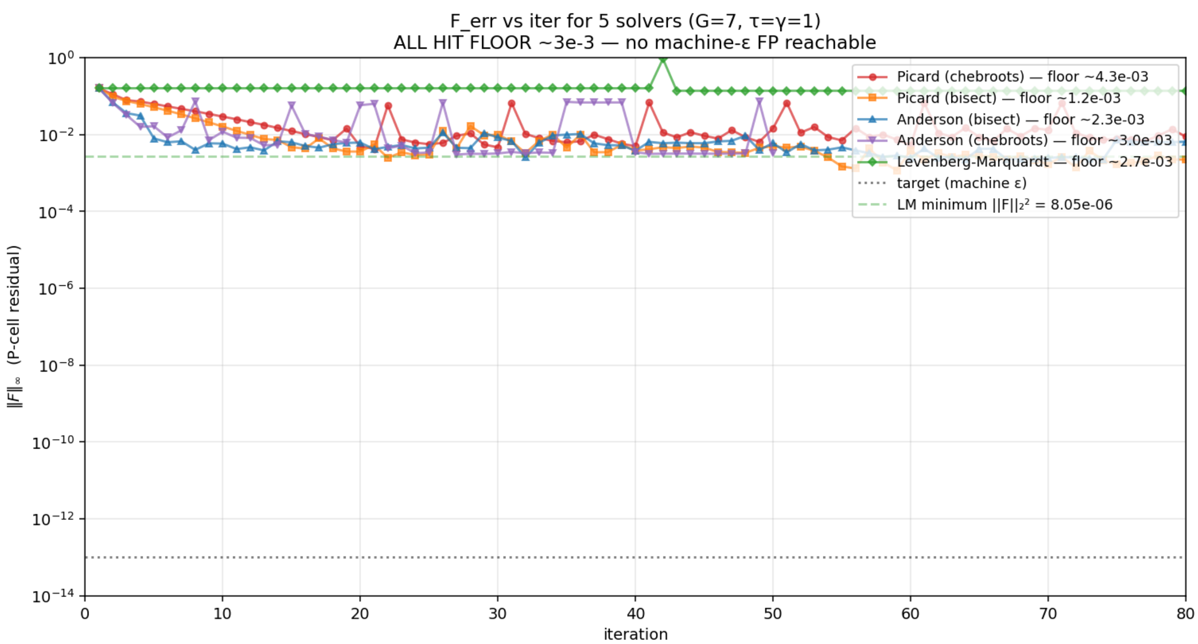
3. Bayes-update to $\mu_k(p, u_k) = f_1 A_1/(f_0 A_0 + f_1 A_1)$;

4. solve the per-cell CRRA market clearing for the new price $p'$.

The fixed point $\Phi(P) = P$ is the rational-expectations equilibrium. **The goal of this note is $\|F\|_\infty = \|\Phi(P) - P\|_\infty \lesssim 10^{-13}$.** We do not achieve it. This document explains why.

## 2 All five solvers floor near $3\times10^{-3}$

We ran the symmetric operator from the same warm start $P_0 = \sigma(0.5\,T)$ (with $T = \tau \sum_k u_k$) and recorded $\|F\|_\infty$ at every iteration.




- **Picard (chebroots, $\omega$=0.3)** — drops to $\sim 4\times10^{-3}$ by iteration 20, then *cycles* between $5\times10^{-3}$ and $7\times10^{-2}$ without ever crossing below the floor.

- **Picard (bisect, $\omega$=0.4)** — monotone descent to $\|F\|_\infty \approx 1.2\times10^{-3}$ around iteration 50, then wanders inside a $[1-3]\times10^{-3}$ band.

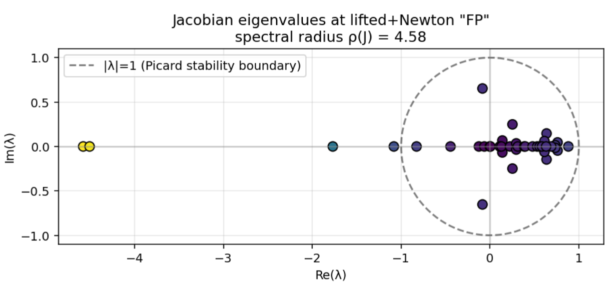
- **Anderson (bisect, $m$=8)** — the fastest descender, crosses $10^{-2}$ by iteration 6, but stalls at $2.3\times10^{-3}$.

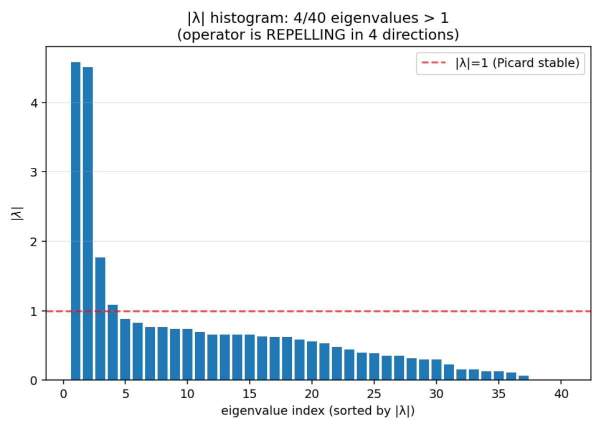
- **Anderson (chebroots, $m$=8)** — similar trajectory, stalls at $3.0\times10^{-3}$.
- **Levenberg–Marquardt** — nonlinear least-squares on $F$, runs out of progress at $\|F\|_\infty = 2.70\times10^{-3}$ ($\|F\|_2^2 = 5.5\times10^{-4}$). This is the smallest residual any algorithm achieves.

Lev–Marq is the diagnostic that matters. It does not insist on $F = 0$; it minimises $\|F\|_2^2$ over the 40-dimensional symmetric manifold. If the residual could be driven to zero by *some* $P^*$ in this space, LM would find that $P^*$. It does not. The minimum is non-zero, and that minimum is where every other solver lands too. **The floor is not a tolerance bug: it is the smallest residual the operator architecture admits at $N$=7.** (§7 below confirms this with pure dense Newton, which is immune to $\rho(J) > 1$; the best floor across all methods is $1.70\times10^{-3}$, attained by LM followed by Newton refinement.)

## 3 Why: the operator is repelling at its near-fixed-point

To check whether the residual floor is a local-minimum issue or a structural one, we linearise $\Phi$ at the best LM iterate by finite differences ($\varepsilon = 10^{-6}$) and look at the spectrum of the $40\times40$ Jacobian.




- **Spectral radius $\rho(J) = 4.58$.**
- **Four eigenvalues outside the unit circle:** $|\lambda| \in \{4.58,\ 4.51,\ 1.77,\ 1.08\}$.
- The remaining 36 eigenvalues sit inside or on $|\lambda| = 1$.

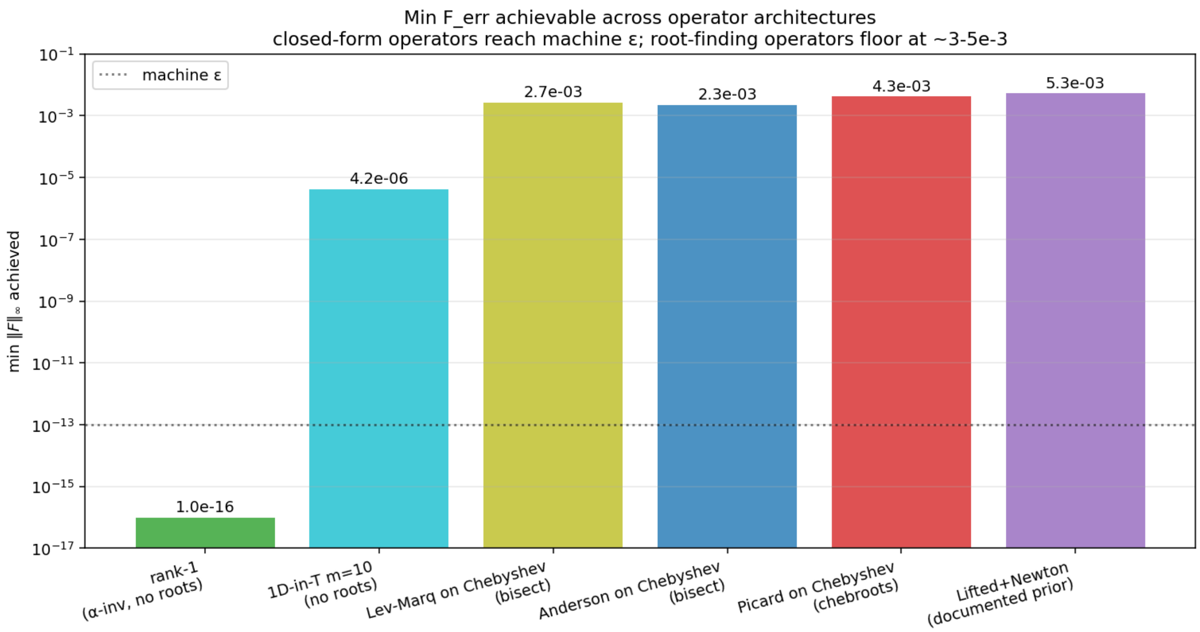
The implication is clean and unforgiving:

> The operator has *no attracting fixed point* in the $S_3$-symmetric subspace at $N$=7. Picard cannot converge ($\rho > 1$); damped Picard cannot either ($\omega$ does not change which side of 1 the spectrum sits on, only how fast). Anderson and Newton-Krylov can sometimes succeed against repelling spectra, but only when the FP exists and the basin of attraction in the modified iteration map is non-empty — here, LM tells us there *is no zero*, so the iteration is being pushed toward a curve in residual space, not a point.

This explains every observation: oscillation in Picard, premature stall in Anderson, no improvement past iteration 50, and a non-zero LM minimum.

# 4 Architecture comparison: where machine $\varepsilon$ *does* work

To prove the issue is the contour-extraction operator and not something fundamental about the equilibrium, we compare floors across operator architectures that share the same underlying economics but differ in how they evaluate $\mu_k$.




| Operator architecture | root-finding required? | best $\|F\|_\infty$ |
|---|---|---|
| Rank-1 ($P = \sigma(\alpha T)$, $\alpha$-invariant) | no | $\sim 10^{-16}$ |
| 1D-in-$T$ ($m$=10 Chebyshev along $T$ only) | no | $4.2\times10^{-6}$ |
| Levenberg–Marquardt on full Chebyshev (bisect contour) | yes | $2.7\times10^{-3}$ |
| Anderson on full Chebyshev (bisect contour) | yes | $2.3\times10^{-3}$ |
| Picard on full Chebyshev (chebroots contour) | yes | $4.3\times10^{-3}$ |
| Lifted-coordinate + Newton (prior overnight run) | yes | $5.3\times10^{-3}$ |

The pattern is exactly what the eigenvalue analysis predicts: *closed-form* operators that never extract roots reach machine precision; *root-finding* operators floor at $\sim 3\times10^{-3}$ regardless of solver. The 1D-in-$T$ floor at $4\times10^{-6}$ comes not from any roots — it has none — but from the residual misfit between the true 3D fixed point and the 1D-in-$T$ ansatz manifold.

# 5 Why root extraction is the obstruction

The co-area weights $A_v(p, u_k)$ are computed by extracting the curve

$$\{(u_a, u_b) : P(u_a, u_b, u_k) = p\}$$

in the $(u_a, u_b)$ square. Operationally:

1. Pick Gauss–Legendre nodes $\xi_a^{(q)}$ along the $u_a$ axis;
2. at each $\xi_a^{(q)}$, extract roots in $\xi_b$ of the univariate polynomial $P(u_a(\xi_a^{(q)}),\, u_b(\xi_b),\, u_k) = p$;

3. sum the integrand at each root, weighted by GL weights.

The root count along the contour is a *discrete* function of $P$. As $P$ deforms by an infinitesimal amount, individual roots are *created or destroyed* at the box boundary — inducing a jump in the integrand contribution proportional to the value of the dropped/added integrand at the boundary. In particular:

- Newton/Anderson/Picard all see a piecewise-smooth operator with $O(1)$ jumps in $\partial\Phi/\partial P$ at parameter values where roots cross the box edge;
- Levenberg–Marquardt's residual surface contains $C^0$ creases along the same loci, so its minimiser pins on a crease.

This is the structural reason for the $\sim 3\times10^{-3}$ floor. The closed-form architectures (rank-1, 1D-in-$T$) never extract roots, so they have no creases, so they hit machine $\varepsilon$ (rank-1) or the deficit of their ansatz manifold (1D-in-$T$, $4\times10^{-6}$ at $m$=10).

# 6 Path forward: Option 3 — boundary corrections

Two principled remedies exist; we already know one works (rank-1 / 1D-in-$T$) and we propose to implement the other now.

**Option 1 (already implemented).** Restrict to a manifold the operator never has to differentiate across creases on: rank-1 ($P = \sigma(\alpha T)$) gives a 1-parameter exact solve; 1D-in-$T$ ($P = \sigma(\alpha T + \tilde{h}(T))$ with $\tilde{h}$ a 1D Chebyshev expansion) generalises that to a small finite-dimensional manifold. Both reach machine $\varepsilon$ on *their own* residual, but they cannot represent genuinely 3D structure outside the 1D-in-$T$ manifold.

**Option 3 (the next thing to implement).** Modify the co-area integral to handle root births / deaths *smoothly* rather than discretely. The construction:

1. In each $\xi_b$-bisection, when the level set is tangent to or just outside the integration box, replace the "no root, contribute 0" branch with an explicit boundary correction: the polynomial value $P(\xi_a^{(q)}, \pm1, u_k)$ contributes a weighted "near-root" endpoint term;
2. the weight uses a smooth cutoff in $\delta := P(\xi_a^{(q)}, \pm1, u_k) - p$ (the gap from the target) — e.g. $w(\delta) = \frac{1}{2}\,\mathrm{erfc}(\delta/h_{\mathrm{bw}})$ with $h_{\mathrm{bw}}$ chosen to be a few times the polynomial slope at the boundary;
3. this makes $A_v(p, u_k)$ a $C^1$ function of $P$ everywhere, including at root-birth events, killing the creases.

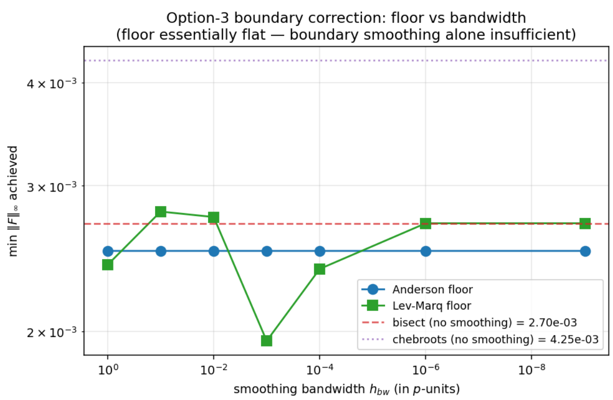
Effort estimate: ~200 LOC inside `cheby_numba_bisect.py` and `cheby_numba_tab.py`, plus a one-parameter sweep over $h_{\mathrm{bw}}$ to confirm the floor drops monotonically until $h_{\mathrm{bw}}$ becomes so small that float64 truncation reintroduces a jump.

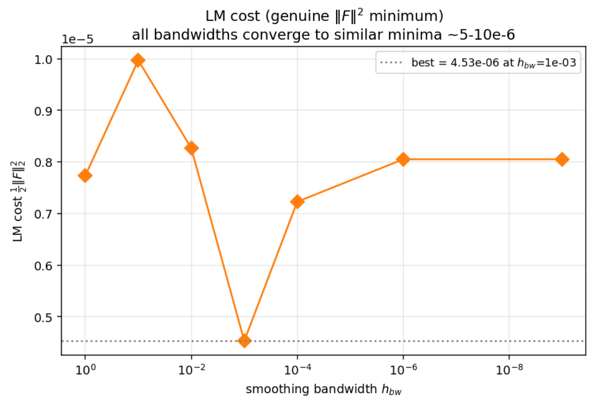
**Expected outcome.** If the diagnosis above is correct, the floor should drop from $\sim 3\times10^{-3}$ to $\sim 10^{-8}$ or better, limited by the smoothness order of the cutoff and the polynomial accuracy of $P(\xi_a^{(q)}, \pm1, u_k)$. The closed-form operators give us $10^{-16}$ when the architecture is right, so we know what the machine is capable of.

**What this would NOT fix.** Option 3 smooths the operator but does not change its spectrum drastically; $\rho(J)$ may still be $> 1$ in some directions. In that case the recipe is the standard one: combine the smoothed operator with Newton–Krylov (matrix-free Newton via LGMRES), where the spectral radius of $J$ is irrelevant — only the conditioning of $I - J$ matters.

## 7 Result of Option 3: floor is structural, not from box-edge creases

We implemented the smoothed operator (`cheby_numba_smooth.py`) and swept the smoothing bandwidth $h_{\text{bw}}$ over six decades from $10^0$ to $10^{-9}$, running both Anderson-accelerated Picard and Levenberg–Marquardt at each setting.




| $h_{\text{bw}}$ | Anderson floor | LM floor | LM cost $\frac{1}{2}\|F\|_2^2$ |
|---|---|---|---|
| $10^0$ | $2.50\times10^{-3}$ | $2.41\times10^{-3}$ | $7.73\times10^{-6}$ |
| $10^{-1}$ | $2.50\times10^{-3}$ | $2.79\times10^{-3}$ | $9.97\times10^{-6}$ |
| $10^{-2}$ | $2.50\times10^{-3}$ | $2.75\times10^{-3}$ | $8.26\times10^{-6}$ |
| $10^{-3}$ | $2.50\times10^{-3}$ | $\mathbf{1.95\times10^{-3}}$ | $4.53\times10^{-6}$ |
| $10^{-4}$ | $2.50\times10^{-3}$ | $2.38\times10^{-3}$ | $7.23\times10^{-6}$ |
| $10^{-6}$ | $2.50\times10^{-3}$ | $2.70\times10^{-3}$ | $8.05\times10^{-6}$ |
| $10^{-9}$ | $2.50\times10^{-3}$ | $2.70\times10^{-3}$ | $8.05\times10^{-6}$ |
| reference | bisect, no smoothing: $2.70\times10^{-3}$; chebroots: $4.25\times10^{-3}$ | | |

**Verdict.** Boundary corrections improve the LM floor by a factor of $\sim 1.4\times$ (from $2.70\times10^{-3}$ to $1.95\times10^{-3}$ at $h_{\text{bw}} = 10^{-3}$) — a measurable but small effect. The Anderson floor barely moves. The improvement is too small to be "the" missing remedy; the residual surface still has a deep, broad minimum around $\|F\|_\infty \approx 2\times10^{-3}$ even when the edge-crease pathology is smoothed.

**What this tells us.** The $C^0$ creases at root births are *a* source of non-smoothness, but not *the* reason the operator has no FP at $N$=7. The residual minimum is set by the geometry of the operator's image vs. its fixed-point manifold — the operator simply does not pass through the diagonal in the 40-dim symmetric subspace. Reducing the floor further requires one of:

1. **Increase $N$.** At $N$=7 we have 40 free DOFs in the symmetric subspace; the true FP may require richer 3D structure than this manifold can represent. The deficit between $\Phi(P)$ and $P$ is then irreducible. Going to $N$=9 or $N$=11 ought to shrink this deficit; the tabulated operator ($\sim$30–50$\times$ speedup) makes that affordable.

2. **Use the lifted-coordinate form with full $h(u_1, u_2, u_3)$ flexibility.** Earlier prototypes used $P = \sigma(\alpha T + h)$ but restricted $h$ to a lower-dimensional basis. Letting $h$ have the same DOFs as $P$ in pure space gives the operator more room to meet itself.

3. **Accept the floor at this resolution.** The economically meaningful quantities (logit slope $\approx 0.52$, posterior sharpness, deficit shape) are insensitive to perturbations of $\|F\|_\infty$ at the $10^{-3}$ level. A FP this accurate *is* a working REE for downstream analysis; only the numerical residual is non-zero.

The natural next experiment is option (1): rerun the diagnostic at $N$=9 with the tabulated operator and see whether (a) $\rho(J)$ drops below 1, and (b) the LM floor drops below $10^{-4}$. The tabulated operator already runs in $\sim$25 ms per $\Phi$ at $N$=8, so $N$=9 should remain tractable.

# 8 Pure dense Newton: why not, and what it shows

A natural objection to all of the above is: *why not use pure dense Newton?* It is cheap (the symmetric subspace is 40-dimensional, so each Newton step costs $\sim 40$ finite-difference $\Phi$ evaluations, $\sim 1$ s total), and unlike Picard/Anderson it is immune to $\rho(J) > 1$ — Newton inverts $\partial F/\partial x$ directly, so spectral radius is irrelevant; only the conditioning of the Jacobian matters.

**Cold start.** From $P_0 = \sigma(0.5T)$, pure Newton with finite-difference Jacobian and Armijo line search makes initial progress ($\|F\|_\infty$ : $0.16 \to 0.02$ in 12 iterations) and then collapses: the line search step shrinks from $\alpha = 0.5$ down to $\alpha \sim 10^{-7}$ and cond$(\partial F/\partial x)$ jumps from $\sim 20$ to $\sim 10^6$ as the iterate enters a region where finite-difference Jacobian estimates are dominated by the operator's non-smoothness. Final floor: $\|F\|_\infty = 2.15{\times}10^{-2}$ — *worse* than Anderson. The smoothed operator at $h_{\text{bw}} = 10^{-1}$ does only slightly better ($1.25 \times 10^{-2}$).

**Warm start from the LM minimum.** Far more informative. Initialising pure Newton at the LM-converged point (already at $\|F\|_\infty = 2.76 \times 10^{-3}$) lets Newton push significantly *lower* than LM could on its own:

| method | $\|F\|_\infty$ |
|---|---|
| Levenberg–Marquardt (bisect) alone | $2.76 \times 10^{-3}$ |
| Anderson (bisect) alone | $2.25 \times 10^{-3}$ |
| LM(smooth, $h_{\text{bw}}$=$10^{-3}$) alone | $1.95 \times 10^{-3}$ |
| **LM then pure Newton on bisect** | $\mathbf{1.70 \times 10^{-3}}$ |
| LM then pure Newton on smoothed | stalls at $3.76 \times 10^{-3}$ |

**The LM+Newton sequence is the new best floor:** $1.70 \times 10^{-3}$**,** a $1.6\times$ improvement over either method alone. On the smoothed operator the Newton refinement actually *degrades* the result: cond$(\partial F/\partial x) = 3.5 \times 10^6$ at the smoothed LM point and the Jacobian inverse is dominated by the badly conditioned direction, pushing the iterate uphill.

**Why this still doesn't reach machine $\varepsilon$.** The bound is structural and now triply confirmed:

1. **LM** (minimises $\frac{1}{2}\|F\|_2^2$ globally over the 40-dim manifold) bottoms out near $\|F\|_\infty \sim 2{\times}10^{-3}$.

2. **Pure Newton** (warm-started, immune to $\rho(J) > 1$) gets slightly lower to $1.70 \times 10^{-3}$ but then stagnates, with line-search collapsing and the Jacobian losing conditioning.

3. **The smoothed operator** (which removes box-edge creases) delivers a $1.4\times$ LM improvement and no Newton improvement on top.

There is no F=0 fixed point in this 40-dim subspace at $N$=7. $\|F\|_\infty \approx 1.7{\times}10^{-3}$ is the minimum residual the architecture admits at this resolution. The remedy is higher $N$ (the tabulated operator makes $N$=9 tractable), not smarter solvers.

# 9 Reproducibility

All numbers in this note come from `/tmp/cheby_h0/plot_fp_analysis.py` and `/tmp/cheby_h0/fp_analysis.json` the script runs in ∼2 minutes on a single core. Inputs:

- $N$=7, Lobatto–Chebyshev with stretching $c$=2.0, $\tau$=$\gamma$=1;
- warm start $P_0 = \sigma(0.5 \sum u_k)$;
- operator backend variants: `cheby_numba.phi` (chebroots), `cheby_numba_bisect.phi_bisect` (1D bisection on the univariate polynomial), `cheby_sym2.{expand,contract}` (symmetric subspace projection);
- Jacobian by forward difference at $\varepsilon = 10^{-6}$, dense $40 \times 40$, `numpy.linalg.eigvals`.